

Koi 8×8: A low-cost 64-channel current-source-and-sense controller for thermo-optic phase shifters in photonic integrated circuits

Rohan Yadgirkar* Ali Elshaari

KTH Royal Institute of Technology, Stockholm, Sweden

rohany@kth.se elshaari@kth.se

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Abstract

This article describes the development of *Koi 8×8*, a compact, low-cost, open-source 64-channel current-source-and-sense controller for driving arrays of thermo-optic (TiN) phase shifters on photonic integrated circuits. The system is built entirely from off-the-shelf parts: a Raspberry Pi Pico controls up to eight stackable daughterboards, each combining an 8-channel DAC, a 24-bit $\Sigma\Delta$ ADC, and eight precision industrial current transmitters operated as current sources. Each channel sources a programmed current and reads back the load voltage, so the host computes heater resistance and dissipated power in situ for closed-loop tuning. On each board the DAC and ADC share one precision reference, so the reference error cancels in the derived resistance; a one-time calibration against a reference source-measure unit recovers absolute accuracy. Complete hardware, firmware, host software, and calibration tooling are released under open licenses.

Specifications table

Hardware name	Koi 8×8 — 64-channel current-source-and-sense controller
Subject area	• Engineering and materials science • Optics and photonics (<i>select final categories from the HardwareX list</i>)
Hardware type	• Electrical engineering and computer science • Measurement / instrumentation and control
Closest commercial analog	Multichannel source-measure systems for photonic biasing, e.g. Qontrol Q8iv/BP modules and multi-channel Keithley 2400-class SMUs. [TODO: confirm exact comparison products + prices]
Open source license	Hardware: CERN-OHL-S v2; Firmware/software: MIT (<i>confirm</i>)
Cost of hardware	[TODO: total build cost from BOM; per-channel cost]
Source file repository	[TODO: Zenodo DOI] (mirror: [TODO: GitHub URL])

1 Hardware in context

Programmable photonic integrated circuits (PICs) have become a central platform for optical communications, sensing, microwave photonics, and increasingly for optical computing and quantum

*Corresponding author: rohany@kth.se

information processing [1]. What makes these circuits *programmable* is a dense fabric of tunable elements — meshes of Mach–Zehnder interferometers and ring resonators whose phase is set electrically [1, 2]. The dominant tuning mechanism is the thermo-optic effect: a resistive micro-heater, commonly a titanium-nitride (TiN) filament patterned over the waveguide, dissipates electrical power and locally shifts the refractive index [3]. A single modern device can integrate from tens to several hundred such heaters, and each must be held at an independent, stable bias for the circuit to hold a programmed state.

Two requirements make driving these heaters non-trivial at scale. First, the relevant control variable is dissipated *power*, not drive current, and the heater resistance drifts with temperature and ages over time; robust operation therefore benefits from reading back the voltage across each heater so that resistance and power can be computed and servoed in situ, rather than driven open-loop [4]. (*verify this closed-loop cite fits; alternatives: Padmaraju & Bergman thermal-stabilization review.*) Second, the channel count is large. Driving every heater from a laboratory source-measure unit (SMU) is accurate but prohibitively expensive and bulky beyond a handful of channels. Dedicated solutions exist at both ends of the spectrum: commercial multichannel driver modules (e.g. Qontrol [5, 6]) offer dense drive with optional readback but at a substantial per-channel cost, while research-grade custom CMOS controllers [7] demonstrate excellent density and power but require ASIC fabrication and are not reproducible open hardware. On the open-source side, source-measure designs are typically single-channel bench instruments (e.g. the uSMU [8]); to our knowledge no published open design combines dense multichannel current drive with per-channel voltage readback at the scale photonic meshes need. **[TODO: sweep HardwareX/JOSS/OSF for further open multichannel drivers before submission — reviewers will check that prior open hardware is cited.]** There remains a need for a low-cost, fully open, and readily reproducible controller that combines dense current drive with per-channel readback.

This paper presents *Koi* 8×8 , an open-source controller named for its stack topology — eight stackable daughterboards of eight channels each, not any particular photonic mesh geometry — that drives up to **64 independent current-source channels** and measures the voltage developed across each load, so the host can compute per-channel resistance and power in situ. The design targets the accuracy regime that matters for heater tuning: *relative / ratiometric* accuracy. Each daughterboard carries its own precision voltage reference (eight in total), shared by that board’s DAC and ADC, so the commanded drive and the measured readback are scaled by the same reference and its error cancels in the derived resistance; channels on the same board are additionally compared against the same reference, so common-mode errors cancel between adjacent tuning elements. Board-to-board matching is bounded by the reference initial tolerance and is characterized in Section 7. **[TODO: report measured board-to-board reference spread (REF5030, $\pm 0.05\%$ initial tolerance) alongside the calibration data.]** Absolute accuracy is available through an optional one-time calibration against a reference SMU.

Contribution.

- A stackable controller + daughterboard architecture scaling from 8 to 64 channels from commodity parts (RP2040, DAC80508, AD7193, XTR200).
- Per-channel closed-loop readout enabling in-situ R and power measurement, with an analytic correction for the on-board sense divider current.
- A fast continuous-conversion scan that reads the full 64-channel array in **0.46 s** (measured, Section 7.7), gated by filter settling rather than by a per-channel restart penalty.
- Open hardware, firmware, host software, and calibration tooling.

2 Hardware description

[TODO: Figure: system block diagram (controller + 8 daughterboards, SPI buses, decoders). Place as figures/block_diagram.pdf.]

The system is a host-commanded instrument. A PC sends per-channel current setpoints (in mA) over USB serial; the firmware sources the commanded current through each load and reports the measured voltage, from which the host computes resistance and power.

2.1 Signal chain (per channel)

Each of the 64 channels is an independent loop: an 8-channel DAC sets the input of a precision industrial current transmitter (Texas Instruments XTR200, a 3-wire current/voltage transmitter) operated as a current source, the sourced current flows through the external load (a TiN heater in the target application) to ground, and an 8-channel 24-bit $\Sigma\Delta$ ADC measures the voltage developed across the load through a 6:1 input divider.

Using an industrial transmitter outside its headline 4–20 mA niche is a deliberate design choice, and it is what makes the channel density affordable: the XTR200 integrates the complete precision output stage in one commodity package ($\sim$ US\$1.5), absorbs the sourcing dissipation in a part designed for continuous industrial drive, and its datasheet figures — span error $\leq \pm 0.065\%$, nonlinearity $\pm 0.003\%$, output impedance $47\text{ G}\Omega$ — match or beat a discrete op-amp current source that would cost more board area, calibration effort, and thermal management per channel. Eight sources per daughterboard therefore add no exotic parts and no heatsinking. The design’s economy comes not from a new circuit but from recognizing that a mass-manufactured industrial part — cheap because it is produced in volume for a market far larger than laboratory instrumentation — already integrates the entire precision source channel; the contribution is the system this enables.

The XTR200’s internals are themselves the de-facto standard current-transmitter architecture — a two-op-amp, sense-resistor-feedback voltage-to-current converter: an input amplifier servos the control voltage across a precision set resistor, and a second amplifier with an integrated PMOS pass transistor reproduces the resulting current at the output. An early revision of this design implemented the same topology discretely with a precision dual op amp (OPA2192) per channel. The integrated part displaced it on three grounds that compound at 64 channels: (i) its transfer function is set by internally trimmed thin-film resistors, so the transconductance is uniform across channels and calibration collapses to a single per-channel offset-current reading (Section 7), whereas a discrete channel’s gain inherits the tolerance of its external resistors and would require a two-parameter fit on every channel; (ii) its $47\text{ G}\Omega$ output impedance renders the source effectively ideal, leaving the sense divider’s known $120\text{ k}\Omega$ shunt as the only load-dependent error — which is what permits the analytic correction in the measurement model below; and (iii) the output stage, short-circuit protection, and active-low output-disable pin (used here for power-up sequencing of the per-board front-end enables) arrive integrated, where a discrete channel would add a pass transistor, protection network, and enable gating per channel — board area that would break the eight-channels-per-daughterboard density. *(verify the exact datasheet numbers + wording against the XTR200 datasheet before submission.)*

2.2 Architecture

A single RP2040 (Raspberry Pi Pico) controller drives up to eight identical daughterboards. Each daughterboard carries one **DAC80508** (8-ch, write-only), one **AD7193** (8-ch ADC), eight **XTR200** current sources, and its own **REF5030** 3.0 V precision reference shared by that board’s DAC and ADC — the basis of the per-board ratiometric accuracy argument (Section 1). Addressing uses two 74HC138 3-to-8 decoders (one per ADC/DAC bank) for chip-select, and one SN74LV595 shift register to gate each board’s analog front ends.

- **SPI0** (shared): AD7193 ADCs + the SN74LV595 front-end enable register. Mode 3, 1 MHz.
- **SPI1** (write-only): DAC80508 bus.

- **Addressing:** shared address lines select daughterboard k via the inverted decoder mapping (address = $8 - k$); one bank enable (ADC_EN / DAC_EN) is asserted at a time.

[**TODO: Pull the full subsystem prose from CLAUDE.md: HC138 selector, AD7193 continuous-scan readout, DAC80508 external-reference/gain setup, XTR595 active-low front-end enable. This is mostly transcription + figures.**]

2.3 Measurement model

The reported voltage is the raw ADC-pin voltage. The true load voltage is recovered with the 6:1 sense divider: $V_{\text{heater}} = V_{\text{ADC}} \times 6$. The divider ($100\text{ k}\Omega + 20\text{ k}\Omega = 120\text{ k}\Omega$) draws a small, *known* current from the load node, corrected analytically:

$$I_{\text{heater}} = I_{\text{source}} - \frac{V_{\text{heater}}}{120\text{ k}\Omega}.$$

This correction is exact and computed from the live measurement (not a calibrated parameter). The XTR200 transconductance ($\approx 0.47\text{ V/mA}$ seed) and a per-channel offset current (I_{OS}) are captured by the calibration in Section 7. The $\times 6$ relation above holds above a low-signal knee at $V_{\text{node}} \approx 44\text{ mV}$, below which the sense path departs from it; the measured transfer function, the additive correction that applies above the knee, and the resulting bound on the usable low end are given in Section 7.4.

3 Design files summary

[**TODO: Fill once files are archived to Zenodo. Every released file gets a row.**]

Design file name	File type	Open source license	Location of the file
Controller board (schematic + PCB)	KiCad / Gerber	CERN-OHL-S v2	[TODO: repo link]
Daughterboard (schematic + PCB)	KiCad / Gerber	CERN-OHL-S v2	[TODO: repo link]
Firmware (src/, include/)	C++ (PlatformIO)	MIT	[TODO: repo link]
Host GUI (koi_gui.py)	Python	MIT	[TODO: repo link]
Calibration / sweep tooling	Python	MIT	[TODO: repo link]
Bill of materials	CSV / XLSX	CC-BY-4.0	[TODO: repo link]

4 Bill of materials summary

[**TODO: Add unit costs, quantities (note per-board vs per-system), suppliers, and part numbers. Provide the full itemised BOM as a supplementary CSV; summarise the major line items here.**] [**TODO: State the assembly method and its cost explicitly — the DAC80508 and AD7193 are fine-pitch packages, so say whether the reference build is hand reflow (stencil + hot plate/oven) or a PCBA service, and include assembly in the system cost either way. HardwareX judges reproducibility by whether a typical lab can actually build it.**]

Designator / component	Qty	Unit cost (USD)	Total cost (USD)	Supplier	Supplier Part No.	Manufacturer Part No.
Raspberry Pi Pico (RP2040) controller	1	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
DAC80508 (8-ch DAC)	8	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
AD7193 (8-ch 24-bit ADC)	8	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
XTR200 current source	64	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
74HC138 decoder	2	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
SN74LV595 shift register	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
REF5030 3.0 V precision voltage reference (one per daughterboard)	8	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Sense divider resistors (100 k/20 k)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Passives, connectors, stacking headers (consolidated)	—	—	[TODO:]	—	—	—
PCBs (controller + 8 daughterboards)	9	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
PCB assembly (fine-pitch reflow / PCBA service)	9	[TODO:]	[TODO:]	—	—	—
System total			[TODO:]			

Placeholder values — table will be updated with final quantities, unit costs, and supplier/manufacturer part numbers from the released BOM spreadsheet.

5 Build instructions

[TODO: Step-by-step assembly with photos: PCB fabrication notes (production uses ENIG finish — HASL test boards oxidise and intermittently fail detection), population order, stacking the daughterboards, reference connections. Then firmware flashing:]

```
pio run -t upload      # build + flash the RP2040 over USB
pio device monitor     # serial console @ 115200
```

6 Operation instructions

The firmware runs a line-based ASCII command server over USB serial (115200, \n-terminated). Each command returns exactly one reply line. Channels use a flat global index $g = \text{board} \times 8 + \text{channel}$ (0–63); unpopulated channels report **nan**. Populated boards are auto-detected at boot. Every OK reply echoes its command name so the host can confirm a reply belongs to the command it sent, and every firmware-initiated line begins with # (boot banner, calibration progress, **STREAM** dumps, diagnostics), so an unprefixd line is always a direct command reply.

Command	Effect	Reply
*IDN?	identify	KOI,8x8,fw1.1
*RST	all currents 0, front-ends disabled	OK *RST
ISETA i0 ...i63	set all 64 currents (mA, g -order)	OK ISETA
ISSET g mA	set one channel	OK ISSET <g>
MEASA? [mask]	measure all 64 (or masked boards)	v0,...,v63
MEAS? g	measure one channel	<volts>
XTR enmask	bit $b = 1 \rightarrow$ board b front-ends enabled	OK XTR 0x..
AVG n	on-micro averages per measurement (1–64)	OK AVG <n>
RATE fs	AD7193 filter word (speed/noise)	OK RATE <fs>
STREAM ON OFF	periodic # dump for monitoring	OK STREAM ON OFF
RESCAN	re-detect boards at runtime	OK RESCAN active=0x.. new=0x..
PING? b	board liveness (ADC ID + one conversion)	OK PING 1 0

A typical experiment loop is **ISETA ...** (set), settle, **MEASA?** (read). A Tkinter host GUI (`koi_gui.py`) polls **MEASA?** and displays raw and divider-corrected voltage for all 64 channels in an 8×8 grid; `sweep_ch0.py` performs a scripted single-channel current sweep.

7 Validation and characterization

(This is the critical section for review. Plan: calibrate every channel against a Keithley 2400 over 0.25–6 mA, then report the figures below.)

7.1 Calibration method

Bench setup. Every channel is terminated in a known load resistor standing in for the TiN heater: a thin-film 1 kΩ part with $\leq 25\text{--}50$ ppm/K temperature coefficient in a large package (to bound self-heating: 36 mW at 6 mA), whose value is taken from an in-situ four-wire measurement rather than from the part grade. A dedicated calibration jig PCB carries all 64 loads, permanently connected so that every channel remains loaded and driven throughout the campaign. Each load carries Kelvin taps at its resistor pads — a sense pair and a force pair — and a 23-channel bank of four-pole relay stages (paired DPDT signal relays) connects one load’s taps at a time to a 6.5-digit DMM (Keithley 2100), operated in offset-compensated four-wire mode for resistance scans so that any constant residual current through the load cancels; a re-positionable sense-only harness maps the relay bank onto each third of the array in turn, without ever interrupting a drive path. The same taps thus serve two functions: four-wire resistance measurement of every load through the jig itself (repeatable at any time, e.g. warm, to bound drift), and Kelvin voltage sensing during the current sweeps, so every sweep point yields the load voltage directly and the

channel current as

$$I = \frac{V_{\text{DMM}}}{R_{\text{meas}}} + \frac{V_{\text{DMM}}}{120 \text{ k}\Omega},$$

the second term being the known sense-divider current (Section 2). Because the DMM is switched across the loads in parallel — never inserted in series — the current path is never broken: all 64 channels remain loaded and driven throughout the scan, i.e. the device is characterized under representative operating conditions rather than one channel at a time into an otherwise idle board. Mechanical relays are used (open-contact leakage in the pA range) so the jig itself cannot contribute at the microampere level being resolved. A metal-foil transfer-standard resistor and a Keithley 2400 source-measure unit inserted in series on a small number of channels provide an independent cross-check of the voltage-over-resistance method. The jig design files (schematic, PCB, control script) are released with the paper. **[TODO: instrument list with calibration status; uncertainty budget table (DMM V, load R tolerance/TCR/self-heating, divider term → total I uncertainty) vs. the $\pm 4 \mu\text{A}$ I_{OS} being resolved.]**

Sweep protocol and fit. Each channel is swept over 0.25–6 mA in both directions — the linear operating regime; sub-100 μA points are excluded because the source cannot sink current, so readings near zero bottom out rather than extrapolate linearly. A per-channel linear fit of commanded versus measured current yields `calSlope` (the transconductance, expected $\approx 0.47 \text{ V/mA}$ and shared across channels, consistent with the trimmed-thin-film argument of Section 2) and `calOffset` (the per-part offset current I_{OS} , approximately equal to the enabled zero-command reading). The calibration proper therefore uses *only* the 1 k Ω loads; many sweep points serve to verify linearity and average noise, not to fit additional parameters.

Separating current-side from voltage-side errors. A single-load sweep, however many points it contains, carries a structural confound: with a fixed load, $V = IR$ is perfectly correlated with I , so an error term that scales with *current* (XTR200 gain error, true I_{OS} , DAC error) is indistinguishable from one that scales with *voltage* (board leakage, the $V/120 \text{ k}\Omega$ divider correction, ADC or divider gain error) — both produce identical residuals. To break the correlation, one board is re-run with its loads exchanged for $\sim 300 \Omega$ and $\sim 2 \text{ k}\Omega$ parts of the same grade: at the same 3 mA setpoint the load then sees ~ 0.9 , ~ 3 , and $\sim 6 \text{ V}$ across the three load sets — the same current at a $\sim 7\times$ span of voltage. Comparing the apparent per-channel offset across the three loads attributes it: an offset that is invariant with load is genuinely current-side (real I_{OS} , and belongs in `calOffset`), whereas one that scales with V is voltage-side (leakage or divider-correction error) and must *not* be folded into `calOffset` — doing so would be correct only at the calibration load and would mis-correct on a real heater of different resistance. The same re-run validates the analytic divider correction directly, exercises a wider portion of the ADC span, and yields a load-independence figure for the source. This is an attribution measurement, not a three-point calibration: if it confirms the error budget is current-side, the remaining seven boards need only the 1 k Ω dataset.

7.2 Current-mode characterization

The current-drive characterization mirrors the per-parameter methodology of a representative commercial module (Qontrol Q8a [5]) so the two can be compared plot-for-plot; all figures below report *Koi*’s own measured data. Note the differing operating ranges (*Koi* 0–6 mA vs. Q8a 0–100 mA), so resolution/precision are additionally reported as a fraction of full scale. *Preliminary bench data (single board, 2026-06-17)*: XTR200 gain is near-ideal (datasheet span error $\leq \pm 0.065\%$, nonlinearity $\pm 0.003\%$), so `calSlope` is treated as a shared constant; the per-channel offset I_{OS} is the term needing measurement, bench-measured $\approx \pm 4 \mu\text{A}$ (datasheet $\pm 10 \mu\text{A}$ max).

Measured drive accuracy (channel 0, 2026-07-30). A 64-point sweep spanning 0.1–6.383 mA was taken with a Keithley 2100 across a $996.4\ \Omega$ four-wire-measured load, in *randomized* setpoint order so that any residual drift appears as scatter rather than as an apparent gain error. The actual sourced current is recovered as $I = V_{\text{DMM}}/R + V_{\text{DMM}}/120\ \text{k}\Omega$ (load plus the sense divider’s share). Against commanded current,

$$I_{\text{actual}} = 1.00319 I_{\text{cmd}} + 2.43\ \mu\text{A}.$$

That is a +0.319 % *gain* error and an I_{OS} of +2.43 μA , well within the datasheet $\pm 10\ \mu\text{A}$. Uncalibrated, the worst deviation across the range is 22.7 μA . The *linearity* residual after removing gain and offset is 0.070 μA rms, i.e. 0.0011 % of the 6.383 mA full scale — a $320\times$ reduction for a two-parameter fit. The source is therefore excellent and essentially all of the drive error is a single uncalibrated scale factor, confirming the shared-slope / per-channel-offset model above. No compliance limit was reached, with the XTR200 ERRORFLAG clear throughout.

(The load resistance is the most leveraged constant in this section: it converts every measured voltage into a current, so an error in it presents as a gain error and nothing else. An earlier campaign assumed 1005.68 Ω for this same load against a correct four-wire value of 996.4 Ω , and that 0.92 % difference was enough to reverse the sign of the reported gain error. Every raw meter voltage is archived with the released data so that R enters only at analysis time.) **[TODO: Repeat across all 64 channels to publish the per-channel I_{OS} distribution; the above is one channel.]** *(RESOLVED — the uniform $-24\ \mu\text{A}$ static offset seen on an early devkit sweep (all channels, boards 0–2) was never a current-side term. It is measurement-side: the sense path’s low-signal gain deficit (Section 7.4) presents above the knee as a fixed $-3.75\ \text{mV}$ ADC-referred error, which divided by the load reads back as an apparently uniform current offset — hence its being uniform across parts and far above the datasheet I_{OS} maximum. Per-channel I_{OS} numbers taken after correcting the sense path are therefore trustworthy; the drive side never had this error.)*

Current transfer function — channel 0, Keithley 2100 across a 996.40 Ω load

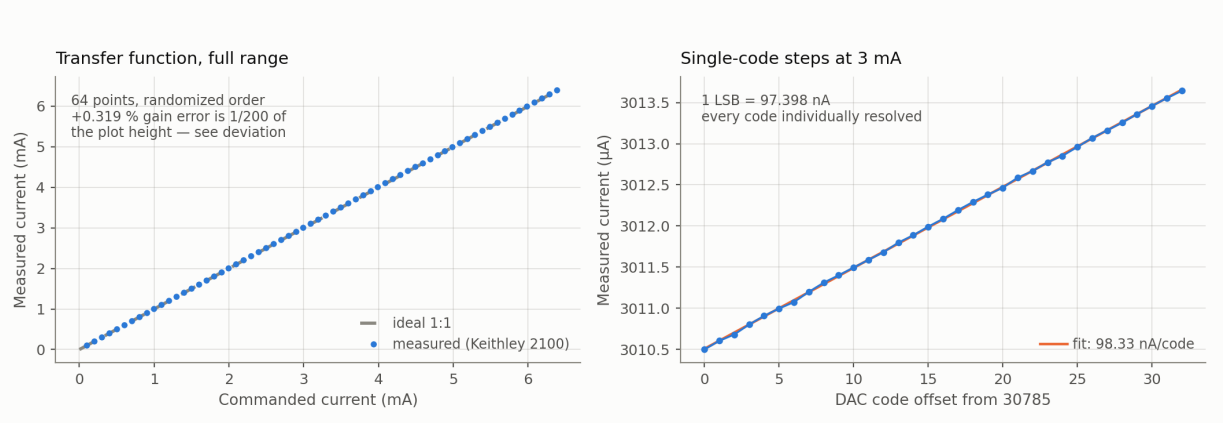


Figure 1: Current transfer function, channel 0. Left: commanded current vs. current measured by the external reference over the full 0–6.383 mA range, 64 points in randomized order; the +0.319 % gain error is far too small to see at this scale, which is why the deviation plot of Figure 2 is the informative view. Right: the single-code region at 3 mA, showing that individual DAC codes (1 LSB = 97.4 nA) are cleanly resolved by the reference.

Code-level resolution and linearity. Resolution was measured with the external DMM alone — the on-board ADC takes no part — by stepping the DAC one code at a time and deriving current from the voltage across the four-wire load. The DAC is 16-bit over the 0–6.3830 mA span, so the instrument offers **65 536 individually settable currents per channel**, on each of the

Current deviation from nominal — source accuracy vs. on-board readback accuracy, channel 0

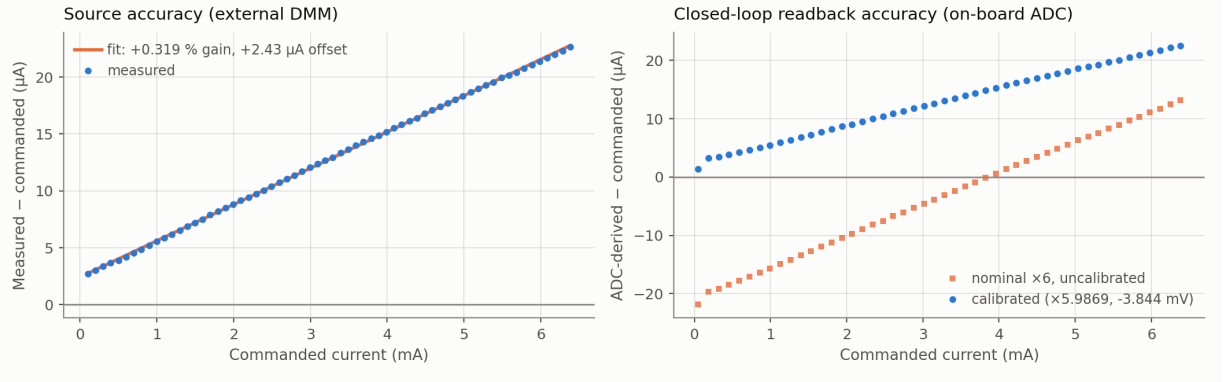


Figure 2: Current deviation from nominal, channel 0. Left: against the external reference DMM — source accuracy. The entire error is one straight line, i.e. a gain term plus an offset, with $0.070\,\mu\text{A}$ rms left over. Right: against the on-board AD7193 readback — closed-loop measurement accuracy, after the sense-divider correction, shown both with the nominal $\times 6$ divider ratio and after the two-parameter sense calibration of Section 7.4.

64 channels independently, with

$$1\text{ LSB} = 97.398\text{ nA} = 97.03\,\mu\text{V across } 996.4\,\Omega.$$

The 2100’s own noise over the same reads is $\approx 8\,\mu\text{V}$, i.e. 0.08 LSB , so the reference resolves roughly twelve steps within a single DAC code and *every code is individually resolvable*; the DAC’s LSB, not the measurement, sets the resolution. Points taken: 64 across the full range in randomized order, plus 33-code ($\pm 16\text{ LSB}$) sweeps centred on 1.0, 3.0 and 5.0 mA. Because a ± 16 -code span is short enough that the converter’s own integral nonlinearity dominates its local slope, the mean step measured that way (98.9 nA) characterizes *step uniformity*, not gain; the gain figure is the full-range fit above.

Quantity	Measured	As % FS
Settable levels	65 536 per channel ($1\text{ LSB} = 97.398\text{ nA}$)	—
INL (randomized order)	rms 0.64 LSB, max -1.31 LSB	0.0020 %
INL (monotonic order)	rms 0.60 LSB, max -1.51 LSB	0.0023 %
DNL	max 0.40 LSB	—
Monotonicity	0 non-monotonic steps in 96	—
Effective resolution	15.6 of 16 bits	—

The converter is monotonic everywhere measured, at every centre and every settling time tried.

Load self-heating, and why it no longer confounds the INL. Measuring INL on a resistive load is not straightforward, because code order and thermal state are confounded: a monotonic 0–6.38 mA ramp raises dissipation from 0 to 40 mW across the sweep, and a metal-film resistor at $\sim 50\text{ ppm}/^\circ\text{C}$ warming a few degrees drifts $\sim 250\text{ ppm}$ — comparable to the $\sim 0.002\text{ % FS}$ being resolved. In an earlier configuration of this bench, with the load convecting freely, a monotonic ramp returned an INL some 60 % larger than a randomized sweep of the same converter, the difference being entirely thermal.

The measurements reported here remove that mechanism at its source: the load is held at ambient by forced air, so its dissipation no longer maps onto code order. Two independent controls confirm it rather than assume it. First, the codes are still visited in random order with

a fixed reference point re-measured every fourth point and each datum corrected by the reference interpolated to its timestamp. Second — and this is the decisive one — the same 48 codes were also swept *monotonically*, the worst case for a self-heating artifact. The two agree: **rms 0.64 LSB randomized against 0.60 LSB monotonic**, well inside the 0.09 LSB per-code repeatability. A thermal term large enough to matter cannot survive that comparison, so the residual is genuine converter nonlinearity.

The practical consequence for anyone reproducing this work is that randomized order is a cheap check rather than a requirement, provided the load is temperature-stabilized; without stabilization it is mandatory, and a monotonic ramp will overstate INL substantially.

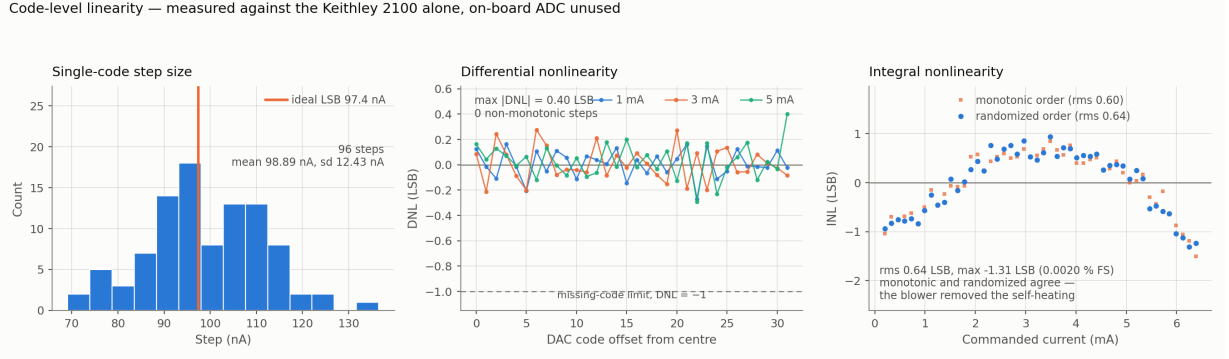


Figure 3: Code-level linearity, channel 0, measured against the external DMM with the on-board ADC unused. Left: single-code step-size distribution (1 LSB = 97.4 nA over the 0–6.383 mA span). Centre: current-mode differential nonlinearity, max 0.40 LSB, with no step approaching the –1 LSB missing-code limit at any of the three centres. Right: integral nonlinearity in randomized and in monotonic code order. With the load held at ambient by forced air the two agree (rms 0.64 vs. 0.60 LSB), which is the evidence that load self-heating no longer contaminates the measurement.

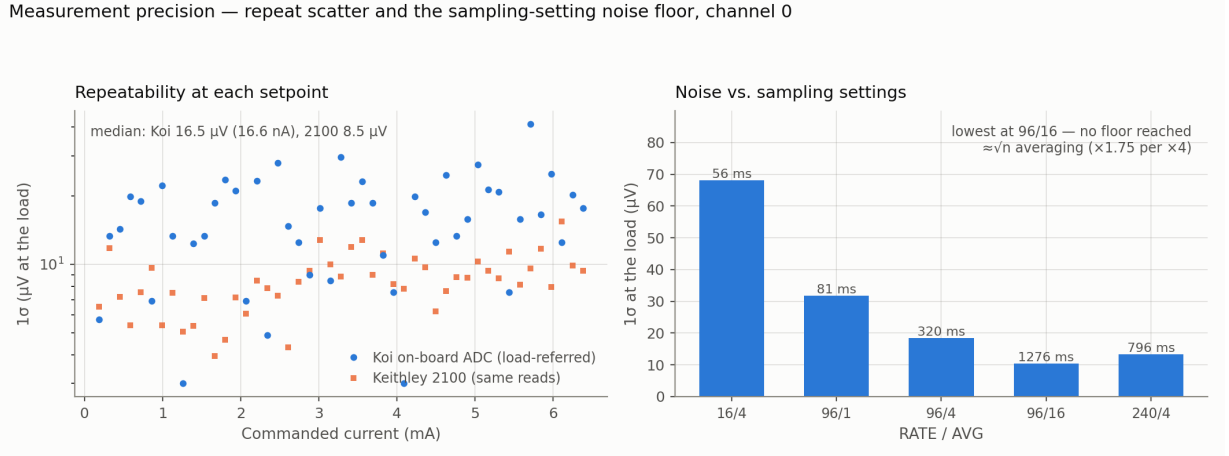


Figure 4: Measurement precision, channel 0. Left: 1σ over repeated reads at each setpoint across the operating range, for the on-board ADC (load-referred) and for the reference DMM over the same reads. Right: noise against the two sampling controls, annotated with the per-read cost. Noise falls monotonically as $\sqrt{n}$ with averaging over the whole range tested, so no floor is reached; the choice of setting is a speed/noise trade rather than a search for a floor (Section 7.5).

7.3 Sense-divider correction and end-to-end resistance accuracy

The divider-current correction $I_{\text{heater}} = I_{\text{source}} - V_{\text{heater}}/120\text{ k}\Omega$ was validated end-to-end by comparing the resistance the instrument reports against the four-wire value of the reference load (996.4Ω). Resistance is computed from the calibrated sense model of Section 7.4, the commanded current, and the divider correction:

Range	R reported	vs. four-wire
$I \geq 0.5\text{ mA}$	$1000.65 \pm 0.80\Omega$	+0.426 %
$I \geq 1.0\text{ mA}$	$1000.44 \pm 0.45\Omega$	+0.406 %
$I \geq 2.0\text{ mA}$	$1000.27 \pm 0.24\Omega$	+0.389 %

The +0.4 % bias is *not* a failure of the divider correction: it is the +0.319 % drive gain error of Section 7.2 propagating through, because R is computed from *commanded* rather than measured current. The quantity that characterizes the instrument is the scatter, $\pm 0.24\text{--}0.45\Omega$ (0.02–0.05 %), which is what remains once `calSlope` is trimmed. Correspondingly, the resistance accuracy of an uncalibrated unit is set by the drive scale factor, and of a calibrated unit by the sense residual (Section 7.4) and the drive linearity (0.0011 % FS).

(This measurement is a useful check on the whole analysis chain. An earlier campaign, with the load resistance mistakenly taken as 1005.68Ω instead of 996.4Ω , reported $-0.372/-0.393/-0.412\%$ with $\pm 0.87/\pm 0.49/\pm 0.27\Omega$ scatter — the same magnitudes and the same scatter as above, with the sign reversed, which is exactly what a 0.92 % error in R predicts. The scatter, the figure of merit here, is unchanged by the correction, as it must be: it does not depend on R .)

7.4 Voltage-sense linearity and the low-signal gain deficit

The voltage-sense path was characterized against a Keithley 2100 6.5-digit DMM connected directly across the load. The DMM was automated over USB-TMC (`tools/keithley2100.py`, driven by `tools/vsweep_dmm.py`), so each sweep point is the mean of eight readings at 10 power-line cycles. Channel 0 was driven into a 996.4Ω resistor (four-wire measured) over $0.005\text{--}3\text{ mA}$, with the ADC at `RATE96`, `AVG4`, gain 1, `SINC4`, chopper off. The DMM was held on a *fixed* 10 V DC range for the whole sweep: autoranging would split the points across two range calibrations and fold the difference between them directly into the fitted intercept, which is the quantity of interest here.

The transfer function is **piecewise linear**, with a knee that this sweep brackets between $V_{\text{node}} = 42.0$ and 46.9 mV — i.e. $44 \pm 2\text{ mV}$, $\approx 44\mu\text{A}$ into this load. The incremental ratio collapses from 10.8 to 7.2 across that single step and reaches its asymptote within two more:

I_{cmd} (mA)	V_{node} (mV)	V_{ADC} (mV)	incr. ratio	implied offset (mV)
0.005	7.245	0.473	—	−0.737
0.010	12.266	0.888	12.09	−1.160
0.020	22.120	1.715	11.82	−1.979
0.030	32.066	2.562	11.64	−2.792
0.040	42.008	3.456	10.80	−3.559
0.045	46.946	4.145	7.17	−3.695
0.050	51.839	4.943	6.13	−3.714
0.100	101.484	13.197	6.009	−3.750
0.200	200.564	29.707	6.001	−3.786
1.000	993.703	162.128	5.988	−3.813
2.000	1985.119	327.735	5.987	−3.765
3.000	2976.507	493.342	5.986	−3.712

Least-squares fits to the two regimes:

$$V_{\text{ADC}} = \underbrace{\frac{V_{\text{node}}}{11.696} - 0.165 \text{ mV}}_{\text{below the knee, } r^2=0.9995} \quad \quad \quad V_{\text{ADC}} = \underbrace{\frac{V_{\text{node}}}{5.9883} - 3.752 \text{ mV}}_{\text{above the knee, } r^2=1.0000000}$$

Interpretation. A residual of $\approx -3.5 \text{ mV}$ at the ADC pin ($\approx -21 \text{ mV}$ load-referred) had previously been characterized as a fixed additive measurement offset. This sweep shows that description is an artifact of never sampling below 0.05 mA : the low-current regime has an intercept of -0.165 mV , i.e. *essentially zero offset*, and instead roughly half the correct gain (11.70 vs. 5.99). The path therefore arrives at the knee already $\approx 3.6 \text{ mV}$ short and thereafter tracks with the *correct* slope, which is indistinguishable from a fixed offset to any measurement taken above the knee. The offset is a consequence of the low-signal gain deficit, not an independent term.

Quantitatively this is consistent with a current drawn from the ADC pin that is ohmic ($\approx 17.4 \text{ k}\Omega$) below $\approx 3.9 \text{ mV}$ and saturates at $\approx 225 \text{ nA}$ above it ($225 \text{ nA} \times (100 \text{ k}\Omega \parallel 20 \text{ k}\Omega) = 16.67 \text{ k}\Omega$) = 3.75 mV) — the resistive-then-constant-current signature of a leaky junction or ESD structure. Note that 225 nA is roughly $200\times$ the AD7193’s buffered input-leakage specification, and that a power-off resistance check cannot detect it (the divider measures a textbook $16.691 \text{ k}\Omega$ pin-to-ground).

Practical consequence. Above the knee — i.e. for any load current above $\approx 45 \mu\text{A}$, which covers the entire intended thermo-optic operating range — the sense path is linear to $r^2 = 1.0000000$, and what remains after the two-parameter fit is $38.6 \mu\text{V}$ rms ($61.5 \mu\text{V}$ worst) at the ADC pin, equivalently $368 \mu\text{V}$ node-referred or **58 ppm of the 6.4 V full-scale node span**. The offset is a single additive constant that calibrates out. Below the knee the reported voltage is *not* trustworthy and should not be used for resistance or power estimation. This bounds the usable low end of the measurement range and is reported here as a characterized limitation.

On the divider ratio. The in-situ fit returns 5.99 rather than the 6.0017 obtained by back-driving the divider with the current source removed. The difference is not a divider error: $\approx 1.9 \Omega$ of series resistance in the load path between the divider tap and the DMM connection carries the load current, so the divider sees $V_{\text{DMM}} + IR_s$, a current-proportional term that lands entirely in the fitted ratio ($6.0017/(1 + R_s/996.4) = 5.988$). The nominal $\times 6$ of Section 2 is correct. The corollary is methodological: an in-situ voltage-vs-voltage sweep cannot measure the true divider ratio and will always return ≈ 5.99 ; ratio must come from a back-drive measurement, and in-situ sweeps should be used for offset and linearity only.

[TODO: Repeat on a second channel and a second board. The data above is one channel on one board; the earlier per-board clustering of this residual makes a 225 nA leak on *every* channel implausible unless it is inherent to the AD7193 input structure, which the single-channel data cannot distinguish.] [TODO: Decisive localization: probe the ADC pin directly across the same low-current points. If the pin voltage itself shows the $11.73 \rightarrow 5.99$ knee the current is drawn at the node (board/ESD leakage, fixable upstream); if the pin tracks a clean 6:1 and only the reported code shows the knee, it is internal to the AD7193. Attach the meter for *all* points so its loading of this unfiltered $\approx 16.7 \text{ k}\Omega$ node stays common-mode.]

7.5 Noise and resolution

Measurement noise was characterized by repeated single-channel reads at a fixed commanded current (channel 0, 1.0 mA into 996.4Ω , 30 reads per setting, the first five after each change discarded because conversions taken before the filter settles are invalid), sweeping the two sampling

Voltage-sense linearity — the low-signal gain deficit that presents as a fixed offset above the knee

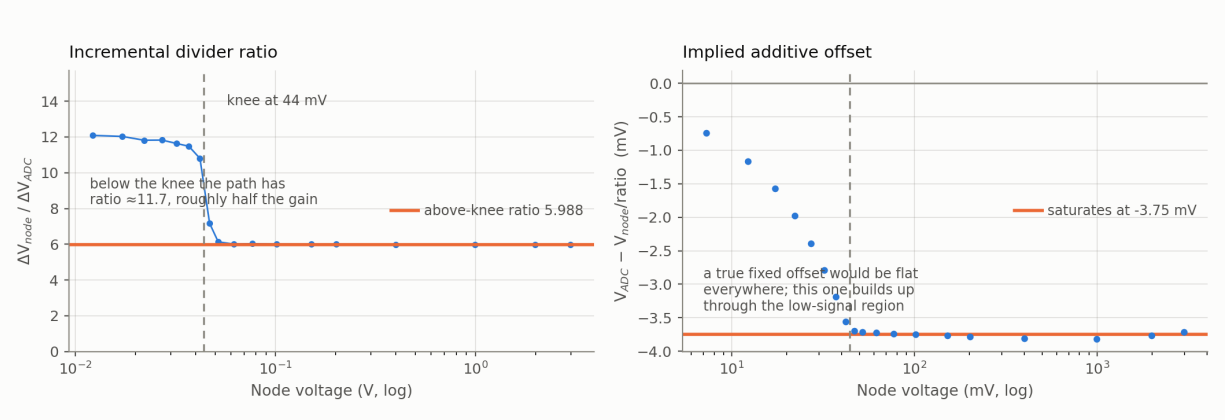


Figure 5: Voltage-sense linearity, channel 0. Left: the incremental divider ratio $\Delta V_{\text{node}}/\Delta V_{\text{ADC}}$ against node voltage, showing the collapse from ≈ 11.7 to the correct 5.99 at the knee ($V_{\text{node}} = 44 \pm 2 \text{ mV}$, dashed). Right: the additive offset implied if one insists on describing the path with the above-knee gain — it builds up through the low-signal region and saturates at -3.75 mV , which is the signature that distinguishes a low-signal gain deficit from a true fixed offset. A genuine offset would be flat everywhere.

controls: **RATE** (the AD7193 filter word) and **AVG** (on-microcontroller averaging). The Keithley 2100's own spread over the same reads is quoted alongside as a bound on how much of the observed noise originates in the source and load rather than the ADC.

RATE	AVG	t/read	1σ at pin	1σ at load	equiv. current	2100 1σ
8	4	29 ms	30.5 μV	183 μV	184 nA	—
16	4	56 ms	11.36 μV	68.2 μV	68.4 nA	8.82 μV
96	1	81 ms	5.29 μV	31.8 μV	31.9 nA	6.29 μV
96	4	320 ms	3.08 μV	18.5 μV	18.6 nA	10.45 μV
96	16	1276 ms	1.72 μV	10.3 μV	10.4 nA	11.94 μV
240	4	796 ms	2.20 μV	13.2 μV	13.3 nA	7.92 μV

Per-read time follows $0.83 \text{ ms} \times \text{RATE} \times \text{AVG}$ at every point measured. Noise falls monotonically across the whole grid — $11.36 \rightarrow 5.29 \rightarrow 3.08 \rightarrow 1.72 \mu\text{V}$, a factor 1.75 per fourfold increase in averaging, i.e. essentially $\sqrt{n}$ — so **no noise floor is reached within the settings tested**, and these data do not establish where one lies. Averaging is also the more efficient lever than filter length here: at $1.6\times$ the time, **RATE 96 / AVG 16** (1276 ms, 1.72 μV) beats **RATE 240 / AVG 4** (796 ms, 2.20 μV). At the quietest setting the pin-referred noise is below the reference DMM's own spread over the same interval, so the figure quoted there is an upper bound on the instrument.

The choice of operating point is therefore a straightforward speed/noise trade rather than a search for a floor. **RATE 8 / AVG 4** refreshes the whole 64-channel array in 1.7 s at 184 nA equivalent noise and is the practical setting for live monitoring; **RATE 96 / AVG 16** is $\sim 18\times$ quieter for deliberate single-channel measurements.

7.6 Thermal validation

Section 2 claims the design needs no heatsinking because the dissipation is distributed: worst case per transmitter is roughly $(V_{\text{supply}} - V_{\text{load}}) \times I \approx 120 \text{ mW}$ (6 mA into a near-short), i.e. $\sim 1 \text{ W}$ per fully driven board spread across eight packages — the same distributed-dissipation approach used by commercial multichannel drivers, so this is standard practice to be *validated*, not a novelty to be claimed. **[TODO: Validate: all 64 channels at maximum current into worst-case (low-R) loads, thermal soak to steady state; report package / board temperatures]**

(IR camera or thermocouple), the temperature rise above ambient, and — the part users actually care about — how much the warm-up shifts the sourced current and the reference (couples to the drift and warm-up tests in the noise section). State the derating, if any, for enclosed / stacked operation.]

7.7 Throughput

The continuous-conversion sequencer reads one full 8-channel board per pass instead of issuing per-channel single conversions, so the readout is gated by filter settling rather than by a fixed per-channel restart penalty. Measured on a fully populated 8-board, 64-channel system:

RATE	AVG	per board (8 ch)	full array (64 ch)	array refresh
8	1	0.064 s	0.46 s	2.17 Hz
8	4	0.224 s	1.74 s	0.57 Hz
16	4	0.436 s	3.43 s	0.29 Hz
32	4	0.860 s	6.82 s	0.15 Hz

All 64 channels returned valid data at every setting. Per-board time $\times 8$ reproduces the full-array time to within 2% throughout, which confirms there is no significant fixed per-board overhead: the scan cost is settling, and it scales exactly as $0.83 \text{ ms} \times \text{RATE} \times \text{AVG}$ per channel.

Two practical notes. The host-side command deadline is 10 s, so whole-array reads above $\approx \text{RATE} \times 48$ at **AVG 4** exceed it and must be issued per board — which is what the host GUI does in normal operation. Second, the readout is not yet optimized: the SPI clock is 1 MHz and the per-board scans are sequential, so overlapping them or raising the clock is available headroom, not a claimed result.

7.8 Channel-to-channel crosstalk and board reproducibility

[TODO: Quantify crosstalk between adjacent channels and consistency across the 8 daughterboards.]



Figure 6: Left: current-mode crosstalk — shift in a victim channel’s current as the other seven channels are swept across their range. Right: power delivered to representative loads vs. commanded current, with the nominal maximum-power point marked.

8 Capabilities and use cases

[TODO: Demonstration: drive an array of TiN heaters on a photonic chip, measure R vs. power, show phase tuning / a programmed mesh state. This demo is what backs the closed-loop-tuning claim in the abstract and Section 1 — it is required, not optional, and is deliberately scheduled *last*: it depends on first resolving the

offset-current anomaly and tightening the measurement chain (Section 7). State limitations: 0–6 mA range, sub-100 μ A unreliable, write-only DAC bus, etc.]

Declarations

[TODO: CRediT author contributions, conflict-of-interest statement, funding, data/code availability (Zenodo DOI), acknowledgements.]

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